

Name \_\_\_\_\_

Period \_\_\_\_\_ Date \_\_\_\_\_

**Practice Test – Advanced Kinematics (Version 6)**

*On any problems, you must show all work, include units, round properly, and box your final answer.*

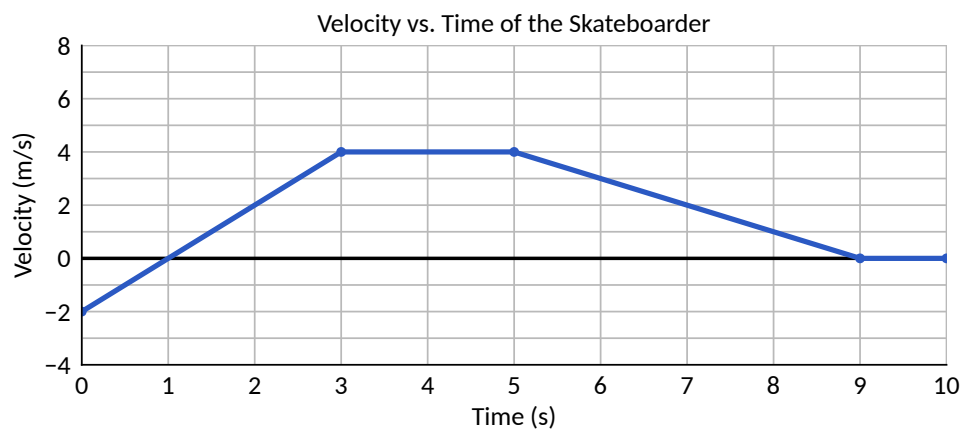
*Do the algebra first, before plugging in values.*

**Rewrite the following statement in the space below:** On my honor, before God, I have neither given, nor received, any aid on this test.

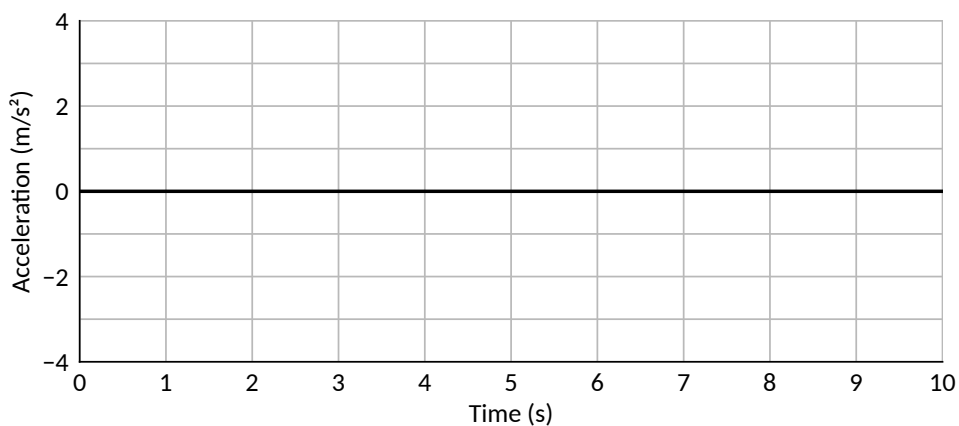
**For all questions:**

- Use  $g = 9.80 \text{ m/s}^2$  and neglect air resistance.
- Round numerical answers to three significant figures.
- 50 minutes. Four questions, 25 points each (100 points total).

**1. (25 points)** The velocity-time graph below shows the motion of a skateboarder rolling along a straight path from  $t = 0$  s to  $t = 10$  s. On the blank grid, draw the corresponding acceleration-time graph. Show the slope calculation for each segment of the velocity-time graph.



Draw your acceleration-time graph here:



Slope calculations:

**2. (25 points)** A train travels along a straight, level track at a constant 27.0 m/s. The engineer applies the brakes, and the train slows with a constant acceleration of magnitude  $1.50 \text{ m/s}^2$  until it stops. How far does the train travel while braking?

**3. (25 points)** Water leaves a lawn sprinkler at ground level with a speed of  $8.00 \text{ m/s}$  at  $60.0^\circ$  above the horizontal and lands on the level lawn at the same height. Assume air resistance is negligible. How far from the sprinkler does the water land?

**4. (25 points)** A rock climber accidentally drops a carabiner from the top of a cliff that is a height  $L$  above the ground below. Assume air resistance is negligible. If down is the positive direction, what is the carabiner's velocity just before it hits the ground? Derive a mathematical expression in terms of  $L$  and  $g$ .

## Answer Key – Practice Test, Version 6 – Advanced Kinematics

Worked step by step, the same way as the class notes: sketch, list the variables, pick the equation, algebra first, then numbers with units.  $g = 9.80 \text{ m/s}^2$ ; answers to three significant figures.

### AP Physics 1 equation sheet, kinematics (Mechanics and Fluids, first three):

#1  $v_x = v_{x0} + a_x t$  · #2  $x = x_0 + v_{x0} t + \frac{1}{2} a_x t^2$  · #3  $v_x^2 = v_{x0}^2 + 2a_x(x - x_0)$

Same equations as  $v_f = v_i + at$ ,  $\Delta x = v_i t + \frac{1}{2} at^2$ ,  $v_f^2 = v_i^2 + 2a\Delta x$ . Pick the one that leaves out the variable you don't know and don't need. For vertical motion use  $y$  in place of  $x$ .

## 1. Acceleration-time graph of the skateboarder (25 points)

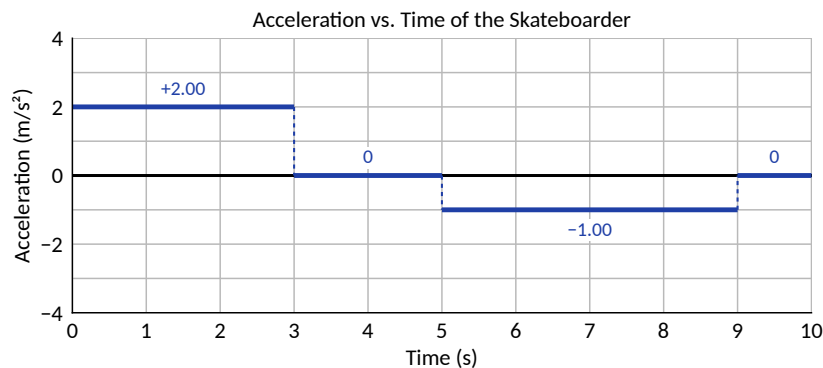
### Step 1 Acceleration is the slope of the v-t graph.

$$a = \frac{\Delta v}{\Delta t} = \frac{v_f - v_i}{t_f - t_i} \text{ (read } v_i \text{ and } v_f \text{ at the start and end of each straight piece; values read off grid lines are treated as exact)}$$

### Step 2 Find the slope of each segment.

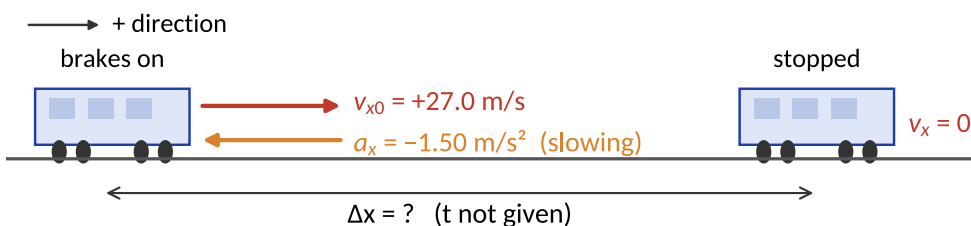
| segment     | $\Delta v = v_f - v_i$                             | $\Delta t = t_f - t_i$                     | $a = \Delta v / \Delta t$    |
|-------------|----------------------------------------------------|--------------------------------------------|------------------------------|
| 0 s to 3 s  | $4 \text{ m/s} - (-2 \text{ m/s}) = 6 \text{ m/s}$ | $3 \text{ s} - 0 \text{ s} = 3 \text{ s}$  | <b>+2.00 m/s<sup>2</sup></b> |
| 3 s to 5 s  | $4 \text{ m/s} - 4 \text{ m/s} = 0 \text{ m/s}$    | $5 \text{ s} - 3 \text{ s} = 2 \text{ s}$  | <b>0 m/s<sup>2</sup></b>     |
| 5 s to 9 s  | $0 \text{ m/s} - 4 \text{ m/s} = -4 \text{ m/s}$   | $9 \text{ s} - 5 \text{ s} = 4 \text{ s}$  | <b>-1.00 m/s<sup>2</sup></b> |
| 9 s to 10 s | $0 \text{ m/s} - 0 \text{ m/s} = 0 \text{ m/s}$    | $10 \text{ s} - 9 \text{ s} = 1 \text{ s}$ | <b>0 m/s<sup>2</sup></b>     |

### Step 3 Draw each slope as a flat line over its time interval. Label both axes with units.



Watch out: from 0 s to 3 s the skateboarder starts out rolling backward ( $v = -2 \text{ m/s}$ ), stops for an instant at 1 s, then rolls forward. The line is straight the whole time, so  $a = +2.00 \text{ m/s}^2$  the whole time, even at the instant  $v = 0$ . Zero velocity does not mean zero acceleration.

## 2. Braking train: stopping distance (25 points)



### Step 1 Pick a positive direction and get the sign of a.

Let the direction the train moves be +. The train is moving + but slowing down, so its velocity is decreasing:  $\Delta v$  is negative, which makes a negative. The slowing acceleration points opposite the motion, so  $a = -1.50 \text{ m/s}^2$ .

### Step 2 List the five variables.

| $v_i (= v_{x0})$ | $v_f (= v_x)$ | $a$                    | $t$                  | $\Delta x$ |
|------------------|---------------|------------------------|----------------------|------------|
| +27.0 m/s        | 0 m/s (stops) | -1.50 m/s <sup>2</sup> | not given, not asked | ? (wanted) |

### Step 3 Pick the equation.

| AP sheet | equation                                   | leaves out                     |
|----------|--------------------------------------------|--------------------------------|
| #1       | $v_x = v_{x0} + a_x t$                     | $x - x_0$ ( $\Delta x$ )       |
| #2       | $x = x_0 + v_{x0} t + \frac{1}{2} a_x t^2$ | $v_x$ (final velocity)         |
| #3       | $v_x^2 = v_{x0}^2 + 2a_x(x - x_0)$         | <b>t (time) ← use this one</b> |

**Why #3:** time is not given and not asked for. #3, written  $v_f^2 = v_i^2 + 2a\Delta x$ , is the only one without  $t$ .

### Step 4 Start from $v_f^2 = v_i^2 + 2a\Delta x$ and set $v_f = 0$ . Solve for $\Delta x$ before any numbers.

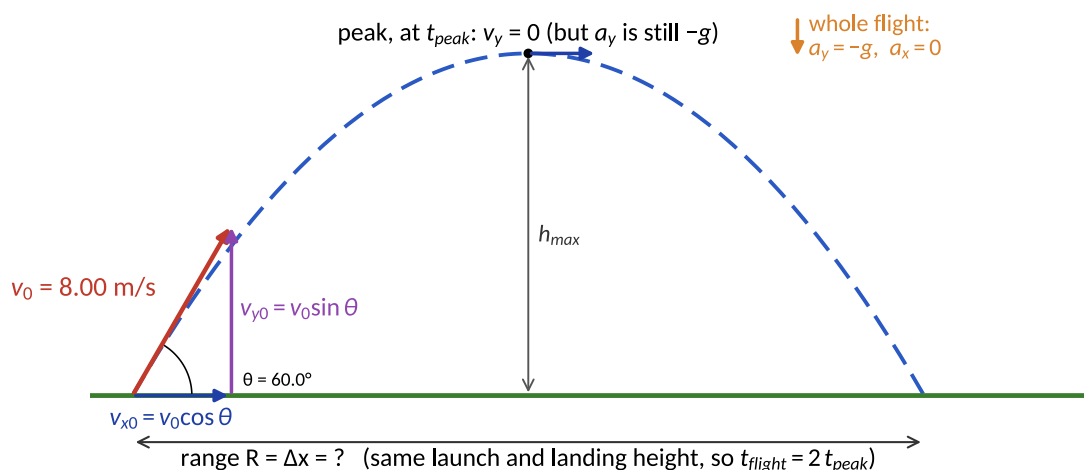
$$v_f^2 = v_i^2 + 2a\Delta x \rightarrow 0 = v_i^2 + 2a\Delta x \rightarrow 2a\Delta x = -v_i^2 \rightarrow \Delta x = -\frac{v_i^2}{2a}$$

### Step 5 Plug in numbers with units.

$$\Delta x = -\frac{(+27.0 \text{ m/s})^2}{2(-1.50 \text{ m/s}^2)} = -\frac{(729 \text{ m}^2/\text{s}^2)}{(-3.00 \text{ m/s}^2)} = \boxed{243 \text{ m}}$$

The minus sign in the formula and the negative  $a$  cancel, so  $\Delta x$  comes out positive: the train moves forward while it stops. If you get a negative distance,  $a$  had the wrong sign. Units:  $(\text{m}^2/\text{s}^2) \div (\text{m}/\text{s}^2) = \text{m}$ .

### 3. Sprinkler: range (25 points)



Split the motion into two motions. Sideways moves at a constant speed ( $a_x = 0$ ). Up and down is free fall ( $a_y = -g$ ). **Time** is the only thing both share.

**Step 1** Resolve the initial velocity into components (keep letters; numbers go in at the end).

$$v_{x0} = v_0 \cos \theta \text{ (the side next to the angle: along the ground)}$$

$$v_{y0} = v_0 \sin \theta \text{ (the side across from the angle: straight up)}$$

**Step 2** Make the T-chart.

| horizontal                 | vertical                         |
|----------------------------|----------------------------------|
| $a_x = 0$                  | $a_y = -g = -9.80 \text{ m/s}^2$ |
| $v_{x0} = v_0 \cos \theta$ | $v_{y0} = v_0 \sin \theta$       |
| $\Delta x = R = ?$         | at the peak: $v_y = 0$           |
| $t = t_{\text{flight}}$    | $t_{\text{peak}} = ?$            |

**Step 3** Find the time to the peak, where  $v_y = 0$ .

| AP sheet | equation                                   | leaves out                                          |
|----------|--------------------------------------------|-----------------------------------------------------|
| #1       | $v_y = v_{y0} + a_y t$                     | $y - y_0 (\Delta y) \leftarrow \text{use this one}$ |
| #2       | $y = y_0 + v_{y0} t + \frac{1}{2} a_y t^2$ | $v_y$ (final velocity)                              |
| #3       | $v_y^2 = v_{y0}^2 + 2 a_y (y - y_0)$       | $t$ (time)                                          |

**Why #1:** at the peak you know  $v_y = 0$ ,  $v_{y0}$  and  $a_y$ , and you want  $t$ . The height of the peak is not known, so use #1, the one that leaves out  $\Delta y$ .

$$v_y = v_{y0} + a_y t \rightarrow 0 = v_0 \sin \theta - g t_{\text{peak}} \rightarrow t_{\text{peak}} = \frac{v_0 \sin \theta}{g}$$

**Step 4 Double it for the whole flight.**

The water lands at the same height it left the sprinkler, so the trip down takes exactly as long as the trip up:  $t_{\text{flight}} = 2t_{\text{peak}} = \frac{2v_0 \sin \theta}{g}$

Watch out: at the peak  $v_y = 0$ , but the acceleration is still  $a_y = -g$ , pointing down. Gravity never switches off. Doubling  $t_{\text{peak}}$  only works because the launch and landing heights match. And  $\Delta y = 0$  over the whole flight means it ends at the height it started, not that the acceleration is zero.

**Step 5 Use the flight time in the horizontal column.**

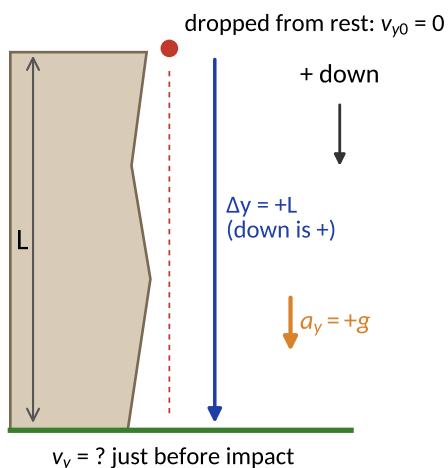
**Why #2 (x-form):** horizontally you know  $t$  and  $v_{x0}$  and want  $\Delta x$ ; with  $a_x = 0$ , #2 becomes  $\Delta x = v_{x0}t$ .

$$x - x_0 = v_{x0}t + \frac{1}{2}a_x t^2 \rightarrow R = v_{x0}t_{\text{flight}} = (v_0 \cos \theta) \frac{2v_0 \sin \theta}{g} \rightarrow \boxed{R = \frac{2v_0^2 \sin \theta \cos \theta}{g}}$$

**Step 6 Now plug in numbers with units.**

$$R = \frac{2(8.00 \text{ m/s})^2(\sin 60.0^\circ)(\cos 60.0^\circ)}{(9.80 \text{ m/s}^2)} = \boxed{5.66 \text{ m}}$$

Check:  $t_{\text{peak}} = (8.00 \text{ m/s})(\sin 60.0^\circ)/(9.80 \text{ m/s}^2) = 0.7070 \text{ s}$ ,  $t_{\text{flight}} = 1.414 \text{ s}$ ,  $v_{x0} = 4.000 \text{ m/s}$ , and  $(4.000 \text{ m/s})(1.414 \text{ s}) = 5.656 \text{ m} \approx 5.66 \text{ m}$ . ✓ (2 sin  $\theta$  cos  $\theta$  = sin  $2\theta$ , so this also equals  $v_0^2 \sin 2\theta / g$ .)

**4. Carabiner dropped off a cliff: impact velocity (25 points), variables only**

No numbers anywhere: the answer is in terms of  $L$  and  $g$ , both positive numbers.

This time **down is +** (the question says so), so the signs flip from the usual setup. The carabiner ends  $L$  below the start, which is the + direction:  $\Delta y = +L$ . Gravity also points down:  $a_y = +g$ .

**Step 1 List the five variables (vertical, so  $y$  in place of  $x$ ).**

| $v_{y0}$ (initial)    | $v_y$ (final) | $a_y$ | $t$                  | $y - y_0 = \Delta y$ |
|-----------------------|---------------|-------|----------------------|----------------------|
| 0 (dropped from rest) | ? (wanted)    | +g    | not given, not asked | +L                   |

**Step 2** Pick the equation.

| AP sheet | equation                                   | leaves out                     |
|----------|--------------------------------------------|--------------------------------|
| #1       | $v_y = v_{y0} + a_y t$                     | $y - y_0$ ( $\Delta y$ )       |
| #2       | $y = y_0 + v_{y0} t + \frac{1}{2} a_y t^2$ | $v_y$ (final velocity)         |
| #3       | $v_y^2 = v_{y0}^2 + 2a_y(y - y_0)$         | <b>t (time) ← use this one</b> |

**Why #3:** t is not given and not asked, and you want the final velocity. #3 is the only one without t.

**Step 3** Solve with letters.

$$v_y^2 = v_{y0}^2 + 2a_y(y - y_0) = 0^2 + 2(+g)(+L) = 2gL \rightarrow v_y = \pm\sqrt{2gL}$$

**Step 4** Pick the root.

It is moving **down** at the ground, and down is + here, so take the positive root:  $v_y = +\sqrt{2gL}$

Speed =  $\sqrt{2gL}$ , same as with up as +; only the sign follows the convention.

Units check:  $\sqrt{(m/s^2)(m)} = \sqrt{m^2/s^2} = m/s$ .

Why not  $-\sqrt{2gL}$ ? With down as +, that would mean moving up at the ground. Don't add a minus just because things fall; the sign comes from the convention.

## Suggested partial credit (25 points per question, 100 total)

### 1. Acceleration-time graph

| Suggested partial credit                                                                                 | pts       |
|----------------------------------------------------------------------------------------------------------|-----------|
| Slope formula $a = \Delta v / \Delta t$ set up with values and units for each of the 4 segments (2 each) | 8         |
| Correct slope value with sign and units for each segment (1 each)                                        | 4         |
| Four horizontal segments at the correct accelerations (2 each)                                           | 8         |
| Changes in acceleration occur at the correct times; no slanted lines                                     | 2         |
| Axes labeled with quantity and units; readable scale                                                     | 3         |
| <b>Total</b>                                                                                             | <b>25</b> |

### 2. Braking train

| Suggested partial credit                                                         | pts       |
|----------------------------------------------------------------------------------|-----------|
| Starts from $v_f^2 = v_i^2 + 2a\Delta x$ (sheet #3) because time is not involved | 5         |
| Sets $v_f = 0$                                                                   | 3         |
| Acceleration negative, with a reason (opposite the motion / slowing down)        | 4         |
| Solves $\Delta x = -v_i^2 / (2a)$ before substituting                            | 6         |
| Substitution with units in every parenthesis                                     | 4         |
| Final answer, 3 significant figures, units, boxed                                | 3         |
| <b>Total</b>                                                                     | <b>25</b> |

### 3. Sprinkler range

| Suggested partial credit                                                           | pts       |
|------------------------------------------------------------------------------------|-----------|
| Resolves $v_0$ into $v_0 \cos \theta$ and $v_0 \sin \theta$                        | 4         |
| Time to peak from $v_y = 0$ with sheet #1: $t_{\text{peak}} = v_0 \sin \theta / g$ | 5         |
| Doubles $t_{\text{peak}}$ for the flight time, with the equal-height reason        | 4         |
| $R = v_{x0}t$ with the expression for $t$ substituted (not a rounded number)       | 5         |
| Substitution with units in every parenthesis                                       | 4         |
| Final answer, 3 significant figures, units, boxed                                  | 3         |
| <b>Total</b>                                                                       | <b>25</b> |

### 4. Dropped carabiner

| Suggested partial credit                                                             | pts       |
|--------------------------------------------------------------------------------------|-----------|
| States $v_{y0} = 0$ , $a_y = +g$ and $\Delta y = +L$ , consistent with down positive | 6         |
| Uses sheet #3 (y-form), since $t$ is not involved                                    | 4         |
| Algebra to $v_y^2 = 2gL$                                                             | 5         |
| Chooses the positive root with a reason (moving down, and down is +)                 | 6         |
| Final $v_y = +\sqrt{2gL}$ boxed, and speed $\sqrt{2gL}$ distinguished                | 4         |
| <b>Total</b>                                                                         | <b>25</b> |